Physics Is Already There: Rethinking Physical Inductive Biases in Latent World Models

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Abstract

Physical state is already in the latent of a world model—injecting it again does not help. Trained without any physics supervision, a JEPA-style latent world model recovers object position by linear probing ($\rho = 0.80$–0.96) and is near-exact for one step (latent cosine 0.98–0.99), yet its rollouts drift away from the dynamics it encodes. The popular remedy is to inject physical inductive biases into the latent. We test this systematically—five mechanism families spanning hard/soft, state/evolution, and labelled/label-free axes (ten arms), across three controlled physics domains and two photorealistic-simulation benchmarks—and find injection fails in 29 of 30 mechanism–domain cells; supplying the *correct* physical form ($a{=}g$), or co-training from scratch, hurts more rather than less ($\Delta{+}0.56$ normalized MSE). We trace this to physical state being *decodable but not load-bearing*: the 190 non-slot dimensions redundantly encode the same position ($\rho = 0.79$–0.92), so prediction routes around any injected slot, and re-weighting that slot restores parity in only 2 of 4 decision cells while never yielding a gain. The real gap is the predictor's long-horizon dynamics, not the latent's state content: repairing the training protocol alone—injecting no physics at all—cuts rollout error 2.2–$8.3\times$. An extrinsic architecture closes the bypass yet still collapses under OOD physics: necessary, but not sufficient.

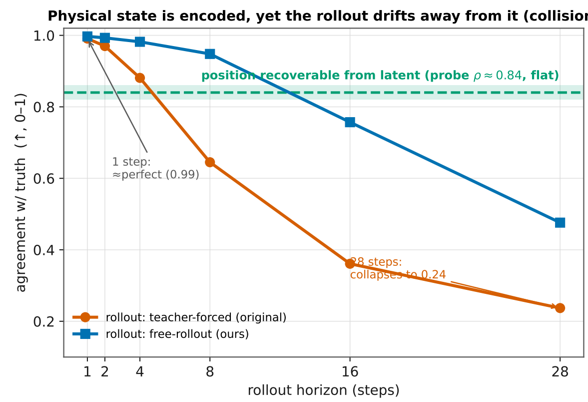

Figure 1: Physical state is encoded, yet the rollout drifts away from it (collision domain). The flat dashed line marks position decodability from ground-truth frame embeddings (probe $\rho \approx 0.84$): the information is *present* at every horizon. The teacher-forced rollout (LeWM's default training) nevertheless collapses from near-perfect one-step agreement (0.99) to 0.24 by horizon 28; training the same model by free rollout (§3.3) stabilizes the long horizon (0.48) without injecting any physics. Presence of state does not imply its use by prediction.

1 Introduction

World models promise agents an internal simulator: given a few frames, they should predict how objects move, fall, and collide (Ha and Schmidhuber 2018; LeCun 2022). Current latent world models fall short of this promise in a specific way—not by failing to *represent* physical state, but by failing to *evolve* it. On the PhyWorld benchmark (Kang et al. 2024), a JEPA-style world model (Maes 2025; Balestriero and LeCun 2025) trained with no physics supervision encodes object position almost perfectly (linear-probe correlation up to 0.96), and its one-step predictions are near-exact (latent cosine 0.98–0.99). Yet when the same model is rolled out autoregressively, prediction error compounds: over 28 steps the latent cosine on collision dynamics falls to 0.24, and out-of-distribution (OOD) physics—unseen radii, masses, or velocities—multiplies rollout error several-fold over the in-distribution level (Figure 1). State decodability does not imply physical-law compliance.

Facing this failure, a natural and increasingly popular remedy is to *inject the physics into the model*. The remedy has real successes—in other model classes: physically interpretable world models route prediction through a dedicated low-dimensional physical state by construction (Mao, Umasudhan, and Ruchkin 2024; Peper et al. 2025), and deep supervision improves prediction in a low-dimensional state latent where the supervised quantities occupy a large fraction of the representation (Zahorodnii 2025). Scaling data does not substitute: Kang et al. (2024) showed that video generators trained at scale still fail OOD physics, generalizing case-by-case rather than by rule. But for *latent* world models—a shared, high-dimensional representation in which prediction happens end-to-end—physical injection has only been tried in isolated, mutually incomparable settings. Whether it works there has never been tested systematically.

That is the question this paper answers: **can injecting**

physics make a latent world model physically consistent? We study it in a controlled setting. A latent world model consists of an encoder mapping frames to embeddings and a predictor rolling embeddings forward, trained by matching predicted to encoded futures (JEPA). *Teacher forcing* trains the predictor on one-step transitions from ground-truth context; *free rollout* trains it on its own multi-step predictions (Bengio et al. 2015; Venkatraman, Hebert, and Bagnell 2015). PhyWorld provides three single-mechanism physics domains—uniform motion, parabolic flight, and elastic collision—with explicit in-distribution ranges over radius/mass and velocity, yielding four evaluation partitions (ID, r/m-OOD, v-OOD, both-OOD). We report normalized MSE (nMSE) in latent space and PSNR of decoded rollouts in pixel space, and use cosine similarity and probe correlation as diagnostics only (§4.1).

Prior evaluations of physical injection leave the basic factorial question open. Injection mechanisms have been tested in isolation—one mechanism, one domain, one initialization regime—and never against the confound that dominates everything else: the training protocol. Prior work treats training choices as background and physical structure as the variable, so no study answers: *does physical structure add anything once the training protocol is done right?* (A secondary hazard, which we treat as a methodological safeguard rather than the main gap: probe correlation and latent cosine are duals of the auxiliary loss and rise mechanically when it is optimized, so they can manufacture apparent gains; verdicts in this paper therefore rest on nMSE and pixel metrics, §4.1.)

We answer the question by exhaustive, controlled anatomy. On a fixed backbone we cross five injection-mechanism families—ten arms spanning hard/soft constraints, state/evolution targets, and labelled/label-free supervision, each with an explicit design rationale (§3.1)—with two training regimes (post-hoc fine-tuning and from-scratch co-training) and three physics domains, then validate the surviving conclusions on photorealistic simulation (Physion and Physion++ (Bear et al. 2021; Tung et al. 2023)). We pair the anatomy with a mechanistic hypothesis—the *load-bearing problem*: physical slots occupy 2 of 192 latent dimensions and $\sim 1\%$ of the prediction loss, while the black-box channel redundantly encodes the same position—and design three falsifiable tests for it (§3.2). As a controlled contrast and positive control, we evaluate the training protocol on identical code and metrics (§3.3); as an external check, we faithfully port an extrinsic physics world model (Mao, Umasudhan, and Ruchkin 2024) onto the same benchmark (§3.4).

Our results separate cleanly. Physical injection fails in 29 of 30 mechanism–domain cells; the damage *grows* as the baseline strengthens (from-scratch co-training costs $+0.56$ nMSE against $+0.035$ post-hoc), and even the exact gravitational form $a{=}g$ hurts ($1.57\times$). The sole seed-robust gain—velocity in a re-weighted slot on parabola—is the exception that proves the mechanism: it helps only where the injected quantity extrapolably drives the dynamics, and reverses sign on the other two domains. All three mechanism tests confirm the load-bearing account: the 190 non-slot dimensions decode position as well as the full latent ($\rho = 0.79$–0.92); the auxiliary term dominates the predictive one by 15–$125\times$ in weighted loss while the encoder's effective dimensionality collapses by 39–90%, ruling out "the constraint never took effect"; and a dose–response re-weighting sweep restores parity in only 2 of 4 decision cells, never a net gain. Meanwhile the training protocol, injecting no physics, cuts OOD rollout error 2.2–$3.6\times$ across all synthetic domains and $8.3\times$ on Physion++, and matching the rollout horizon to dynamics complexity drives long-horizon error to $1/19$ of baseline. The extrinsic port learns the correct physics and beats our model in-distribution, yet collapses where the OOD shift hits its encoder (ρ 0.33 vs. 0.89)—closing the bypass is necessary but not sufficient.

This paper contributes:

- the first systematic, seed-controlled anatomy of physical inductive biases in latent world models—five mechanism families, ten arms, two training regimes, three synthetic domains, two photorealistic-simulation benchmarks—showing injection fails as a general lever ($29/30$; §4.2);
- a mechanistic account, *decodable but not load-bearing*, with three falsifiable tests: a black-box bypass probe, a loss-dominance and effective-dimensionality measurement that rules out implementation failure, and a dose–response re-weighting sweep (§4.3);
- a controlled factorial showing the dominant variable is the training protocol, not physical structure—doubling as the positive control that excludes implementation and metric artifacts (§4.4);
- an architectural attribution: a faithful extrinsic port closes the bypass but still collapses under encoder-OOD shift, making extrinsic design necessary but not sufficient (§4.5).

2 Related Work

Physically structured world models. A growing line of work argues that world models should carry explicit physical state. Physically interpretable world models align a dedicated low-dimensional state with physical quantities and route prediction through it (Mao, Umasudhan, and Ruchkin 2024); design principles for such models are catalogued by Peper et al. (2025); deep supervision injects physical readouts into the training loss and improves prediction in a low-dimensional state latent, where the supervised quantities occupy $\sim 38\%$ of the representation (Zahorodnii 2025); Hamiltonian networks impose conservation structure on the dynamics (Greydanus, Dzamba, and Yosinski 2019). We acknowledge these successes in their own settings, and test the transferable core of the proposals—fixed slots, kinematic equations including the exact gravitational form, probe supervision, consistency losses, and label-free variants—inside a shared-latent JEPA model, where we find all of them unhelpful or harmful (§4.2). Our external port of the extrinsic architecture (§4.5) attributes the discrepancy to architecture rather than to the physics.

Latent world models and physics benchmarks. JEPA-style models predict in representation space (Assran et al. 2023; Bardes et al. 2024; Balestriero and LeCun 2025;

Maes 2025); Dreamer-line models train on imagined rollouts (Ha and Schmidhuber 2018; Hafner et al. 2023). PhyWorld (Kang et al. 2024) generates controlled single-mechanism physics videos with explicit in-distribution ranges and showed that scaled video-diffusion generators memorize cases rather than laws; Physion and Physion++ (Bear et al. 2021; Tung et al. 2023) provide photorealistic—not camera-captured—simulation with rich multi-object dynamics. We use PhyWorld's OOD protocol to dissect *latent* world models rather than pixel-space generators, and validate surviving conclusions on Physion/Physion++.

Rollout training and exposure bias. Training autoregressive predictors on their own outputs is a classic remedy for compounding error: scheduled sampling (Bengio et al. 2015), Data-as-Demonstrator (Venkatraman, Hebert, and Bagnell 2015), and professor forcing (Goyal et al. 2016) all attack the train–test mismatch of teacher forcing. We claim no novelty for free rollout. Its role in this paper is methodological (§3.3): a controlled factorial variable—the confound prior injection studies never controlled—and a positive control that separates "physical structure does not help" from "our implementation or metrics are broken."

3 Method: A Design Space of Physical Injection

We answer the question of §1 by construction: enumerate the ways physics can be injected into a shared latent, state what each design is supposed to achieve, and pre-register the tests that would falsify our mechanistic explanation. Section 4 reports the corresponding results subsection-by-subsection.

3.1 Injection Design Space: Five Families, Ten Arms

We span the design space along three orthogonal axes—*hard vs. soft* constraints, pinning *what the state is* vs. *how it evolves*, and *labelled vs. label-free* supervision—so that any "try a different encoding" objection lands on an axis we have measured. Table 1 lists the ten arms with the design rationale of each.

Orthogonally, we cross injection with the *training regime*: post-hoc fine-tuning of a pretrained encoder vs. from-scratch co-training, a 2×2 per domain. This closes the remaining escape hatch—"structure must be baked in during pretraining, as extrinsic architectures do."

3.2 Mechanistic Hypothesis and Three Falsifiable Tests

Hypothesis (the load-bearing problem). The physical slot occupies 2 of 192 dimensions and receives $\sim1\%$ of the prediction gradient, while the black-box channel *redundantly* encodes the same position—so prediction can route around any injected slot. Injecting state the latent already carries adds no signal, but does consume capacity and gradient. We design three tests, each of which could falsify part of this account:

1. **Bypass probe (probe-190).** Split the latent into the 2 slot dimensions and the 190 black-box dimensions and decode position from each separately, with a random-2-dimension control. If the black box alone cannot decode position, the redundancy claim is false.
2. **Is the constraint acting at all?** Two independent measurements: (a) the ratio of the *weighted* auxiliary loss to the prediction loss—a proxy for gradient magnitude, not a measured gradient norm; (b) the encoder's participation ratio (effective dimensionality). If the ratio is $\approx$1 and the participation ratio unchanged, the constraint never took effect and the failure is an implementation artifact—the "bug hypothesis." If the ratio is $\gg 1$ and dimensionality collapses, the constraint is acting, and competing with prediction.
3. **Dose–response re-weighting.** Sweep the slot's prediction-loss weight from 1 to 300. If harm does not respond systematically to the weight, the gradient-share mechanism is wrong; if it responds but weighting to saturation still cannot rescue every domain, the bypass is architectural rather than a matter of injection strength—and only an architecture that *closes* the bypass could work, a prediction §4.5 tests.

3.3 The Training-Protocol Control: Free Rollout

Free rollout serves two methodological roles; neither is a novelty claim (it is scheduled sampling, Bengio et al. 2015).

Controlled factorial variable. Physical structure has never been compared against the training protocol on the same baseline; prior studies treat the protocol as background. We set the protocol first—replacing teacher forcing (one-step, ground-truth context) with free rollout (H=8 autoregressive steps, supervising the whole rollout)—and then ask whether structure adds incremental value. Teacher forcing hides compounding error at training time (*exposure bias*): the model never sees its own mistakes, then must consume them at deployment. Free rollout exposes them during training, injecting no physics whatsoever. A second protocol variable, the rollout horizon H, must *match dynamics complexity*: domains with late causal events (collision; photorealistic scenes) need horizons long enough to contain them.

Positive control. A negative result invites two deflations—"your implementation is broken" or "your metrics cannot detect improvement." A protocol change that produces large gains on identical code and identical metrics rules out both at once, pinning the failure on the injected structure itself.

3.4 External and Cross-Backbone Controls

Extrinsic port. We faithfully port the physically interpretable world model of Mao, Umasudhan, and Ruchkin (2024)—a three-stage pipeline (VAE encoder, 128-D; a supervised extractor to a low-dimensional physical state; known-equation dynamics with learnable constants)—onto the same PhyWorld domains and OOD protocol. The port tests our mechanism's architectural prediction from the outside: extrinsic designs make the physical state the sole pathway prediction must route through, closing the bypass by construction.

4 Experiments

Each subsection reports the results of the corresponding design in §3: §4.2↔§3.1, §4.3↔§3.2, §4.4↔§3.3, §4.5↔§3.4.

4.1 Setup

Model. LeWM (Maes 2025), a compact JEPA-style action-conditioned world model ($\sim$10M parameters): a ViT-tiny encoder producing a 192-dimensional embedding, an MLP projector, an action encoder, and a 6-layer causal-transformer predictor, trained to minimize latent prediction error plus a SIGReg anti-collapse regularizer (Balestriero and LeCun 2025). Unless stated otherwise, the encoder is initialized from the public PushT checkpoint; §4.2 additionally studies from-scratch training. The compact, fully controlled backbone allows the complete factorial anatomy ($\sim$200 training runs) with headline configurations repeated over three seeds (3072/1234/42); remaining cells are single-seed and marked as such.

Domains and OOD protocol. PhyWorld (Kang et al. 2024): three single-mechanism domains—*uniform motion* (zero acceleration), *parabola* (constant gravity), *collision* (elastic impulse; discontinuous)—each trained on its $1{,}000$ in-distribution trajectories (32 frames, 224×224; ID means radius/mass $\in$ $[0.7, 1.5]$, velocity $\in$ $[1, 4]$) and evaluated on four partitions: ID, r/m-OOD, v-OOD, both-OOD. The small clean training set follows PhyWorld's controlled protocol and isolates the inductive-bias question from data-scale confounds. Photorealistic simulation: Physion (Bear et al. 2021) for zero-shot object-contact-prediction (OCP) transfer, and Physion++ (Tung et al. 2023) for direct training with latent rollout to horizon 64; both are rendered simulation, not camera video.

Metrics and decision rules. Verdicts use two scale-aware, *external* metrics: latent nMSE ($\|\hat{\mathbf{z}}-\mathbf{z}\|^2$ over target variance) and pixel PSNR of decoded rollouts—external because no injection arm optimizes them directly, and they agree in every case we examined. Latent cosine and probe decodability ρ serve as *diagnostics only*: both are duals of the auxiliary training losses and rise mechanically when those losses are optimized, so as verdicts they systematically overstate physical competence (we measured reversals up to cosine $1.50\times$ better while nMSE degrades to $0.21\times$; our own early sweeps were misled this way before re-judging on prediction loss). nMSE has the opposite failure mode: its denominator degenerates when target variance collapses (a parabola ball exiting the frame drives late-horizon nMSE to 10^6 while cosine stays normal), so parabola is judged on its clean r/m-OOD partition and all numbers are reported per partition and per horizon. Full pathology catalogue and protocol checklist: Appendix D. Splits are trajectory-level; three-seed results report mean$\pm$sample std.

4.2 Injection Fails in 29 of 30 Cells (↔§3.1)

Figure 2 and Table 2 answer every design rationale of Table 1, family by family. *Hard state injection*: the slot faithfully absorbs position (its supervision loss falls $2.53 \rightarrow 0.015$; slot decodability rises $0.31 \rightarrow 0.96$), yet prediction worsens in all three domains ($1.40\times$/$1.23\times$/$1.66\times$)—the constraint acts, harmfully. *Soft injection*: the deep-supervision recipe, effective in a low-dimensional state latent where supervised quantities occupy $\sim$38% of the representation (Zahorodnii 2025), does not transfer to a 192-D visual latent where they occupy 2%: all three domains degrade ($1.27\times$/parity-after-seeds/$1.65\times$), and combining soft with hard does not rescue ($1.54\times$ on collision). *Correct physics does not help*: the strict $a{=}g$ head—the ground-truth functional form—degrades uniform by $1.57\times$ and parabola from 0.127 to 0.173, no better than an unconstrained MLP (0.178). *The purpose-built design fails worst on its target*: the consistency loss, designed for collision impulses, is among the worst exactly there ($1.63\times$). *Labels are not the issue*: the label-free prior and its perfectly labelled counterpart fail in the same direction (collision $1.66\times$ vs. $1.33\times$)—the failure is architectural, not informational.

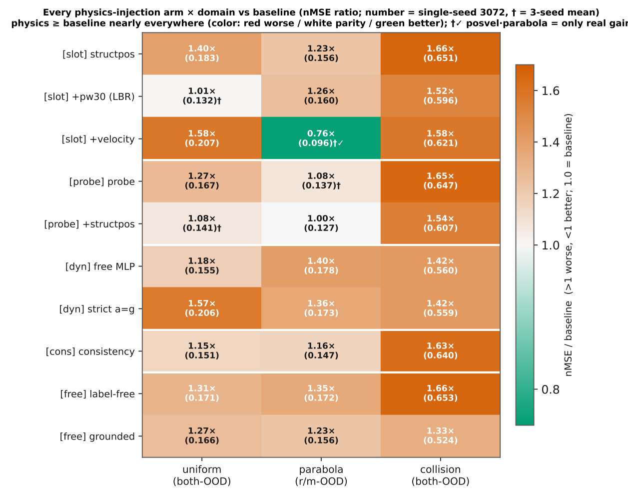

Figure 2: Every injection arm $\times$ domain against the free-rollout baseline (nMSE ratio; number = single-seed value, † = three-seed mean). Red = worse, white = parity, green = better. Injection is at or below baseline in 29 of 30 cells (25 worse, 4 parity); the collision column (1.33–$1.66\times$) is uniformly worst; the sole real gain is velocity+slot on parabola ($0.76\times$, †✓).

The sole exception is the mechanism's signature, not a usable prior: adding *velocity* to the re-weighted slot helps on parabola (r/m-OOD $0.122\pm0.007 \rightarrow 0.096\pm0.007$, every seed lower) because there velocity is an extrapolable driving variable ($a{=}g$ makes it linear in time); the same encoding is the *worst* arm on uniform ($+0.076$, velocity is a redundant constant) and clearly harmful on collision ($+0.025$ over the reweighted slot; velocity jumps discontinuously). A prior that requires knowing the domain's dynamics in advance, and reverses sign across domains, is a diagnostic, not a method—and its best case (-0.026 nMSE) is an order of magnitude below the protocol lever of §4.4.

From-scratch co-training hurts more. The remaining defense—structure must be baked in during pretraining—

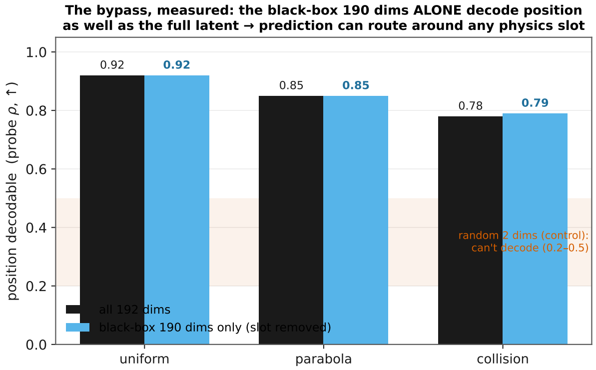

Figure 3: Direct evidence of the bypass. Per domain: position decoded from all 192 dimensions (dark) vs. from the 190 black-box dimensions alone (blue) are nearly equal (0.78–0.92); a random 2-dimension control (band) fails. The black box redundantly encodes position, so prediction need not consult an injected slot.

fails in the strongest way (Table 3): the physics penalty on uniform grows from $+0.035$ (post-hoc) to $+0.558$ (from-scratch); all six regime–domain cells are positive. The cleaner the baseline, the larger the harm.

Two corroborations independent of PhyWorld nMSE. First, *zero-shot transfer* (different dataset, metric, and task): on Physion OCP, no synthetic-trained configuration exceeds the random-encoder prior (0.607 mean AUC); the physics-structured variant is the *worst* of all (0.551), below free rollout (0.603)—"more structure, worse transfer" (Appendix **??**). Second, *photorealistic simulation*: trained directly on Physion++, every injection arm we ported (fixed slot, consistency, consistency+acceleration) degrades per-scenario rollout nMSE by 3–10$\times$ against the identically trained free-rollout model (Table 4); the gaps dwarf the three-seed noise band of the baseline. The probe family has not yet been run on Physion++, so this corroboration currently covers the slot and consistency families.

4.3 Mechanism: Decodable but Not Load-Bearing (↔§3.2)

The three pre-registered tests return the following verdicts.

Test 1: the bypass exists. Probing *only the* 190 *non-slot dimensions* decodes position at $\rho = 0.79$–0.92 across all three domains—as well as the full latent, undiminished by pinning the slot—while a random-2-dimension control fails (0.2–0.5; Figure 3). Position is a redundant, distributed copy the predictor can route through while ignoring any structured slot.

Test 2: the constraint acts—harmfully. The bug hypothesis is ruled out. At probe weight λ=10 the weighted auxiliary-to-prediction loss ratio—our proxy for gradient magnitude (not a measured gradient norm)—reaches 15–125$\times$ across domains, and the encoder's participation ratio collapses by 39–90% (uniform: $41 \to 4$ effective dimensions); at λ=1, the setting of §4.2, the ratios are benign yet

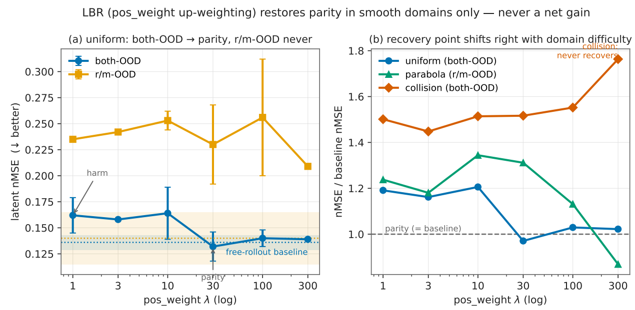

Figure 4: Dose–response re-weighting. Left: uniform, slot weight $1 \to 300$ against the baseline band—both-OOD returns to parity, r/m-OOD never does. Right: nMSE ratio to baseline across domains—collision stays above baseline at every weight and degrades as weight grows. Re-weighting validates the mechanism's direction but is not a repair.

the harm persists. Together with the slot's own decodability rising $0.31 \to 0.96$, the constraint demonstrably reshapes the representation—in a direction that competes with prediction.

Test 3: dose–response holds; rescue does not. Sweeping the slot weight $1 \to 300$ (Figure 4) moves the harm systematically—the mechanism's direction is confirmed—but weighting rescues only 2 of 4 domain–partition decision cells, to *parity* and never a net gain: uniform both-OOD returns exactly to baseline (0.132 ± 0.017 vs. 0.136 ± 0.008), uniform r/m-OOD never recovers, and collision worsens monotonically with weight. The bypass is architectural: no injection strength closes it. This pre-registers the prediction that only an architecture that closes the bypass can help—tested next.

4.4 The Protocol Control: 2.2–8.3$\times$ on Identical Code (↔§3.3)

Replacing teacher forcing with free rollout cuts the hardest-split error 2.2$\times$ (uniform, $0.300{\pm}0.007 \to 0.136{\pm}0.008$), 3.6$\times$ (parabola, $0.443{\pm}0.048 \to 0.122{\pm}0.007$), and 2.4$\times$ (collision, $1.152{\pm}0.048 \to 0.479{\pm}0.079$), with improvements on *every* partition including ID (2.0–4.6$\times$)—it is not an OOD patch (Figure 5). On Physion++ the same switch is worth 8.3$\times$ at horizon 64 ($1.174{\pm}0.041 \to 0.141{\pm}0.018$, three seeds). The horizon must match dynamics complexity: collision improves markedly at $H{\approx}20$ ($0.393 \to 0.294$) while smooth domains degrade with excess horizon (uniform horizon-28 cosine $0.969 \to 0.814$ at H=16); photorealistic scenes have the largest appetite—extending H=$8 \to 28$ (with geometric scale jitter stabilizing amplitude; Appendix C) drives horizon-64 nMSE from 0.280 to 0.014, a $\sim$19$\times$ reduction with no inflection found (Figure 6).

As the positive control of §3.3, this result carries the methodological weight of the paper: on the same code, metrics, seeds, and data that judged all thirty injection cells, a physics-free intervention produces multiples—so the injection failure of §4.2 is attributable to the injected structure, not to the pipeline or the yardstick.

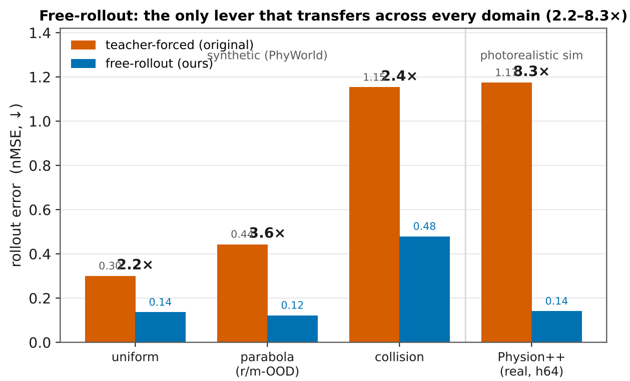

Figure 5: Teacher forcing vs. free rollout (hardest clean split, nMSE ↓; fold-change above bars). One training-protocol switch, injecting no physics: uniform $0.300 \to 0.136$ ($2.2\times$), parabola $0.443 \to 0.122$ ($3.6\times$), collision $1.153 \to 0.479$ ($2.4\times$), Physion++ horizon-64 $1.174 \to 0.141$ ($8.3\times$); all three-seed with non-overlapping intervals.

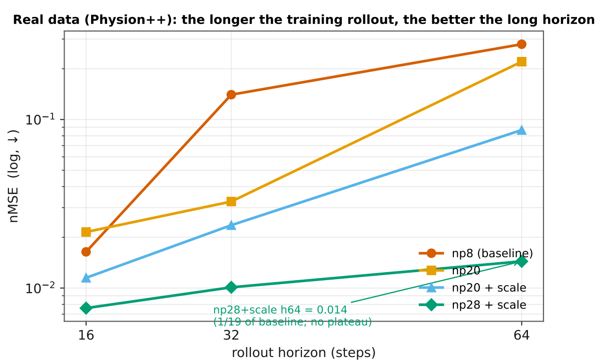

Figure 6: Physion++ direct training: per-horizon rollout nMSE (log scale) as the training horizon grows ($H{=}8 \to 28$, +scale jitter). Long-horizon error falls monotonically to $1/19$ of baseline with no inflection: photorealistic dynamics reward long-rollout training more than the synthetic domains do.

4.5 External Control: Extrinsic Is Necessary, Not Sufficient (↔§3.4)

The faithful extrinsic port learns the correct physics and, where its encoder is in-distribution, beats our shared-latent model: rolled-out position ρ reaches 0.96 (ID) and 0.97 (v-OOD) vs. LeWM's 0.93/0.87 (Table 5). But under radius/mass shift—which changes appearance, hitting the VAE encoder—it collapses to $\rho = 0.33$ (r/m-OOD) and 0.48 (both-OOD) while LeWM holds 0.89/0.87. The architectural prediction of §4.3 is confirmed from the outside: making the physical state load-bearing by construction is what extrinsic designs buy (their equations alone do not transfer, §4.2); yet closing the bypass leaves the encoder-OOD problem unsolved. Extrinsic structure is necessary but not sufficient.

4.6 Synthesis

Same latent, same code, same metrics: injecting state the latent already encodes fails in 29/30 cells (§4.2) because prediction routes around it (§4.3); repairing the long-horizon dynamics of the predictor—with no physics at all—yields 2.2–8.3× (§4.4); and closing the bypass architecturally helps only until the encoder meets OOD physics (§4.5). Physical state in a latent world model is *decodable but not load-bearing*.

5 Conclusion

We asked whether injecting physics can make a latent world model physically consistent, and answered with a controlled anatomy: across five mechanism families, ten arms, two training regimes, three synthetic domains, and two photorealistic-simulation benchmarks, injection fails in 29 of 30 cells—with the exact gravitational form, with perfect labels, and *more* when co-trained from scratch. The reason is that the physics is already there: the shared latent redundantly encodes the state being injected ($\rho = 0.79$–0.92 from the non-slot dimensions alone), so injection adds no signal while consuming capacity and gradient, and prediction routes around it—*decodable but not load-bearing*. All three pre-registered tests support this account, and its two predictions held: re-weighting removes the slot's harm but only to parity, and an extrinsic architecture that closes the bypass by construction does help—until the OOD shift hits its encoder (ρ 0.33 vs. 0.89), making extrinsic design necessary but not sufficient. What actually moved prediction was never the structure: repairing the training protocol—free rollout, and horizons matched to dynamics complexity—delivered 2.2–8.3× on identical code and metrics.

The constructive reading: to help a latent world model, an inductive bias must supply what the latent *lacks*. Its state content is not what it lacks; its long-horizon dynamics are. Injections that target dynamics or conserved quantities rather than state, and extrinsic designs paired with OOD-robust encoders, are the two routes our results leave open—the former untested here beyond the suggestive velocity-on-parabola signature, the latter necessary but demonstrated insufficient alone.

Limitations. Our training-dynamics conclusions rest on one compact backbone (∼10M ViT-tiny JEPA); a frozen-encoder study across seven encoders up to 749M parameters supports the representational conclusions, and replication on a second JEPA backbone is in progress. Headline comparisons, the re-weighting result, and the seed-refuted cells are three-seed; the remaining 26 scan cells are single-seed but sit 5–20 baseline seed-std above baseline. The Physion++ injection arms are single-seed with 3–10× gaps, and the probe family remains untested there. Physion and Physion++ are photorealistic *simulation*, not camera video. The one favorable cell (velocity on parabola) suggests dynamics-matched injection deserves systematic study; we did not perform it.

Reproducibility. All training configurations, per-run logs, evaluation protocols, and the exact numbers behind every table and figure are archived alongside the code; the study builds on public assets (LeWM, PhyWorld, Physion, Physion++). We will release the evaluation suite, including the per-horizon/per-partition audit tools and the random-encoder calibration controls.

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Arm	Design rationale
Family 1: fixed slot (hard · state · labelled)	
slot (structpos)	pin dims [0:2] to true position: the most direct state injection; tests whether a hard constraint helps
+reweight	raise the slot's share of the prediction loss (weight 30): if failure is due to the slot's $\sim 1\%$ gradient share, this should rescue it—falsifiable
+velocity	encode velocity too: tests whether *extrapolability* of the injected quantity is what matters
Family 2: deep-supervision probe (soft · state · labelled)	
probe	linear head reads position from the latent: replicates the deep-supervision recipe (Zahorodnii 2025) in a high-dimensional visual latent
+slot	soft + hard combination: tests complementarity
Family 3: kinematic head (hard · evolution · labelled)	
free MLP	attach $z+v+a$ update with learned a: upgrade from pinning state to pinning *evolution*
strict $a=g$	exact gravitational form, one learnable constant: closes the "your physics was wrong" objection
Family 4: consistency (soft · evolution · labelled)	
consistency	finite-difference velocity of the rolled-out positions must match ground truth; assumes no form for a, designed for collision impulses
Family 5: label-free prior (soft · evolution · unlabelled)	
label-free	slot must evolve as smooth second-order dynamics, no ground truth: the only injection available for raw video
grounded	same structure + true labels: isolates the label variable

Table 1: Ten injection arms grouped into five families, each with its design rationale. The axes close: hard (families 1, 3) vs. soft (2, 4, 5); state (1, 2) vs. evolution (3, 4, 5); labelled (nine arms) vs. label-free (one). All arms run on top of the strongest training protocol (§3.3), which is the deployment-relevant comparison.

	uniform (both-OOD)	parabola (r/m-OOD)	collision (both-OOD)
Free-rollout baseline	0.131	0.127	0.393
Fixed slot	0.183	0.156	0.651
+reweight (w=30)	0.114^{s}	0.160	0.596
+velocity	0.207	$\mathbf{0.096}^{\mathrm{m}}$	0.621
Probe	0.167	0.115^{n}	0.647
+slot+reweight	0.125^{s}	0.127	0.607
Kinematic (free MLP)	0.155	0.178	0.560
Kinematic (strict $a=g$)	0.206	0.173	0.559
Consistency	0.151	0.147	0.57–0.64
Label-free prior	0.171	0.172	0.653
Grounded prior	0.166	0.156	0.525

Table 2: The 30-cell scan (hardest clean split, latent nMSE ↓). [m]The one seed-robust gain: 0.096 ± 0.007 over three seeds vs. baseline 0.122 ± 0.007, non-overlapping. [n]Refuted by seeds: three-seed 0.137 ± 0.034 vs. baseline 0.122 ± 0.007—the single-seed low was seed luck. [s]Also refuted by seeds: reweighted slot 0.132 ± 0.017 and probe+slot 0.142 ± 0.017 vs. baseline 0.136 ± 0.008 (parity). Remaining cells are single-seed but sit 5–20 seed-std above baseline.

Domain	Regime	phys. off	phys. on	Δ
uniform	scratch	0.192	0.750	$+\mathbf{0.558}$
	post-hoc	0.131	0.166	+0.035
parabola (r/m-OOD)	scratch	0.343	0.375	+0.03
	post-hoc	0.124	0.198	+0.07
collision	scratch	0.538	0.635	+0.097
	post-hoc	0.393	0.525	+0.132

Table 3: Pretraining vs. post-hoc injection (latent nMSE ↓, hardest clean split; strict $a=g$ head + fixed slot). Physics hurts in every cell, most where the baseline is cleanest.

Scenario	FR	+slot	+cons.	+cons.+acc.
friction_platform	**0.041**	0.119	0.078	0.067
mass_collision	**0.083**	0.510	0.682	0.696
mass_dominoes	**0.058**	0.452	0.586	0.597
mass_waterpush	**0.122**	0.363	0.663	1.476

Table 4: Physion++ direct training: per-scenario rollout nMSE (↓) at 20 epochs. Injection degrades photorealistic-simulation prediction by 3–10× on the mass/friction scenarios shown (deformable-object scenes are the hardest category for all methods, including the baseline). Single-seed; the gaps far exceed the baseline's seed noise.

Partition (uniform)	Extrinsic port	LeWM (FR)
ID	$\mathbf{0.96}$	0.93
r/m-OOD	0.33	$\mathbf{0.89}$
v-OOD	$\mathbf{0.97}$	0.87
both-OOD	0.48	$\mathbf{0.87}$

Table 5: Rolled-out position decodability ρ ($\uparrow$) of the faithfully ported extrinsic model vs. LeWM with free rollout. The extrinsic design wins where its encoder is in-distribution and collapses where the shift hits the encoder (radius/mass changes appearance): closing the bypass is necessary, not sufficient.

A Training and Evaluation Details

Hyperparameters. All PhyWorld runs: AdamW, lr 5×10^{-5} (from-scratch reruns: 2×10^{-5}), weight decay 10^{-3}, bf16, gradient clip 1.0, 20 epochs (60 for from-scratch cells; the pretraining-regime 2×2 uses 60 epochs uniformly), batch size 64 for free rollout (H=8), 128 for teacher forcing (H=1), 48 for $H\geq16$. History size 3; embedding dimension 192; SIGReg weight 0.09. Seeds: 3072/1234/42 where seed-replicated. Physion++ runs use the readout split (800 clips) with the same optimizer.

OOD partitions. ID: radius/mass $\in[0.7,1.5]$, velocity $\in[1.0,4.0]$. OOD draws radius/mass from $[0.3,0.6]\cup[1.5,2.0]$ and velocity from $[0,0.8]\cup[4.5,6.0]$. Collision uses the four-column variant (two objects' masses and velocities). Training sets are ID-only (1,000 trajectories); evaluation covers all four partitions ($\geq$1,600 trajectories per domain).

Probe protocol. Ridge regression (α=1), trajectory-level 80/20 splits, probing the encoder output (not the projector), pretrained "paper" initialization, and K=4 stacked consecutive latents for velocity. Each choice matters: the naive combination (random init, single frame, through-projector) reads ρ=0.166 for uniform velocity where the corrected protocol reads 0.939 (Trap 4, Appendix D). Random-encoder and pixel-statistics controls are run alongside every probe.

B Full Injection Results

Probe/slot factorial (uniform, free rollout). Latent both-OOD nMSE / pixel both-OOD PSNR (dB): baseline 0.131 / 20.41; probe-only 0.167 / 19.83; slot+reweight 0.114 / 21.30; probe+slot+reweight 0.125 / 20.02. The combination underperforms the slot alone on both scales; the latent and pixel verdicts agree.

Slot reweighting sweep (uniform). Slot weight 1/30/100: 0.183 / 0.114 / 0.136 (single seed), with the w=30 cell seed-replicated at 0.132 ± 0.017 against the baseline 0.136 ± 0.008. Kinematic head on the reweighted slot: velocity-OOD decoded position ρ=0.991/0.987 (x/y); uniform pixel horizon-28 23.34 vs. 22.09 dB (+1.25); appearance-OOD remains the weak axis (decoded ρ 0.29$\to$0.43, vs. 0.96 without the kinematic head).

Bypass probe details. Decoded position ρ per domain (uniform/parabola/collision). Baseline model: full 192 dims 0.92/0.85/0.78; black-box [2:192] alone 0.92/0.85/0.79. Slot-pinned model (structpos, w=30): slot dims [0:2] alone 0.92/0.85/0.51; black-box [2:192] alone 0.92/0.86/0.79—unchanged by pinning. The slot absorbs position (its supervision loss falls 2.53$\to$0.015) without displacing the black box's redundant copy.

Encoded quantity vs. domain (position monotone, velocity sign-flipping). The reweighted slot's effect is systematic in what it encodes (hardest clean split; baseline: uniform both-OOD 0.131, parabola r/m-OOD 0.127, collision both-OOD 0.393):

slot content	uniform	parabola	collision
position (structpos)	0.132	0.160	0.596
position+velocity	0.207	0.096	0.621
Δ (adding velocity)	+0.075	−0.064	+0.025

Position is an output variable with the same role in every domain, so pinning it degrades monotonically with dynamics complexity and never flips sign. The marginal effect of adding velocity (+0.075/−0.064/+0.025) tracks velocity's role in each domain—redundant constant (uniform), linear driver (parabola), discontinuous jump (collision). The parabola gain ($0.122\pm0.007\to0.096\pm0.007$, every seed lower) is thus the sole seed-robust structural gain in the grid: a fixed form landing on a matching domain, not a portable prior.

Physion++ per-scenario (all ten scenarios, FR at 20 epochs). Rollout nMSE: bouncy_platform 0.041, bouncy_wall 0.094, friction_cloth 0.071, friction_collision 0.062, friction_platform 0.041, mass_collision 0.083, mass_dominoes 0.058, mass_waterpush 0.122; deformable scenes are the common hard case for every method (deform_clothhang 1.375, deform_clothhit 0.772). Injection arms degrade the rigid-body scenarios by 3–10× (Table 4) while leaving the deformable failure in place.

Physion++ horizons. Overall nMSE at horizons 16/32/64: baseline (H=8) 0.016/0.140/0.280; H=20 0.022/0.033/0.220; H=16+scale 0.010/0.035/0.136; H=20+scale 0.012/0.024/0.087; H=28+scale reaches horizon-64 nMSE 0.014. Free rollout is the main long-horizon lever (horizon-32 cosine 0.91 vs. 0.79–0.87 for every physics-augmented variant); geometric scale jitter acts as an amplitude stabilizer for long rollouts: without it, H=20 explodes on friction-collision (nMSE 24.6 at cosine 0.99); with it, 0.019.

C Augmentation: a Domain-Specific Lever with a Reversal Boundary

Augmentation is not part of the paper's main line (it is neither an injection mechanism nor a protocol variable), but its boundary behavior is a useful caution. On synthetic domains, per-trial appearance jitter (brightness/contrast, strength 0.5) roughly halves the hardest split: uniform 0.131$\to$0.068 (−48%), parabola −63% (r/m-OOD 0.127$\to$0.065); geometric scale jitter, only in combination with a long horizon, sets the collision record (0.393 $\to$ 0.172 $\pm$ 0.004 over three seeds). Interactions are non-monotone: appearance conflicts with long-horizon training (0.472, worse than either alone), scale synergizes (0.208), the triple is worse than the best pair (0.253); temporal-stride augmentation loses even to appearance on velocity-OOD (0.556 vs. 0.376), leaving unseen velocities the open hard case.

The appearance recipe reverses on photorealistic simulation: the same jitter that halves synthetic OOD error degrades Physion++ friction-collision prediction by two orders of magnitude (nMSE 0.062 $\to$ 6.44) while the *aggregate* long-horizon cosine improves (Trap 1 below), and it does not help zero-shot transfer either (0.597, below the 0.607 random-prior ceiling). The boundary is principled: appearance jitter encodes an appearance-invariance assumption that holds under PhyWorld's simple renderer but breaks where appearance carries physical information (friction, mass, material). Scale jitter, which assumes only size-invariance, keeps helping at photorealistic scale (Appendix B); appearance jitter flips sign. Augmentation gains are domain- and type-specific, not portable.

D Metric Pathologies and Evaluation Traps

Our anatomy repeatedly reached conclusions opposite to those suggested by common metrics; the decision rules of §4.1 distill the following four failure modes, each of which reversed at least one finding.

Trap 1: cosine and probe metrics are duals of the training loss. Probe correlation and latent cosine rise mechanically when probe or slot losses are optimized—they measure whether information is *present*, not whether prediction *uses* it. Probe supervision lifts velocity decodability from ρ=0.44 to 0.80 and gives the most stable long-horizon *cosine* (0.882 vs. 0.843), while its *pixel* rollout is the worst in the comparison (19.93 vs. 20.64 dB at horizon 28). Appearance augmentation on Physion++ improves the aggregate long-horizon cosine while per-scene nMSE collapses up to $100\times$ (deform_clothhit: cosine $0.610 \rightarrow 0.913$, nMSE $0.772 \rightarrow 3.69$). Judging either result by direction-only or readout metrics inverts the conclusion. Our own early sweep was misled this way: judged on K=4 probe ρ, λ=50 "won"; re-judged on prediction loss, the ranking reversed.

Trap 2: nMSE denominators can degenerate. When the target sequence degenerates (a parabola ball exits the frame, target variance $\rightarrow 0$), per-horizon nMSE explodes—3×10^4–2×10^6 at horizon ~ 28 across all six parabola baseline arms, while cosine sits at a healthy 0.55–0.95—and pollutes every aggregate that includes late horizons. This is why parabola is judged on its clean r/m-OOD partition throughout. Traps 1 and 2 point in opposite directions (cosine under-reports failure; degenerate nMSE over-reports it); neither metric family is safe alone, hence per-partition, per-horizon cross-validation.

Trap 3: zero-shot transfer is bounded by the architecture prior. On Physion OCP, a *randomly initialized* encoder scores mean AUC 0.607. No synthetic-training configuration exceeds it: free rollout approaches it (0.603), physics variants fall below it (0.551–0.582), appearance augmentation 0.579–0.597; training longer approaches the ceiling monotonically (0.554 at epoch 1 to 0.603 at epoch 20) without crossing. Per-scenario, only Support ($+0.10$) and Collide ($+0.07$) show signal above the random control. Zero-shot claims must be calibrated against the random-prior floor, or they measure the architecture, not the learning.

Trap 4: protocol confounds fabricate results in both directions. Three probe-protocol choices (random vs. pretrained initialization, single-frame vs. stacked probes, probing through the projector) multiply into a false negative: the naive stack reads uniform velocity decodability at R^2=0.166, the corrected protocol 0.939. A silent initialization bug (only 24 of 216 loadable encoder keys actually loaded) invalidated an entire 45-configuration sweep before a loading guard caught it. And converged training loss proves nothing about rollout: our clean from-scratch baseline converges to the pretrained model's prediction loss (0.006 vs. 0.005) while its rollout OOD error is $2.8\times$ worse.

Auxiliary-loss dominance (measurement behind §4.3, Test 2). At probe weight λ=10, the weighted auxiliary-to-prediction loss ratios—used as a proxy for gradient magnitude; we do not measure gradient norms directly—are $15\times$ (collision), $44\times$ (uniform), and $125\times$ (parabola); the encoder's effective (participation-ratio) dimensionality collapses by 39–90% relative to baseline (uniform $41 \rightarrow 4$). At λ=1 the ratios are benign, which is the setting used in §4.2.

Checklist. (i) Report per-partition *and* per-horizon; never aggregate across horizons without inspecting them. (ii) Judge with scale-aware prediction metrics (nMSE, decoded-pixel PSNR); use cosine and probe ρ as diagnostics only. (iii) Audit nMSE denominators for degenerate targets. (iv) Calibrate transfer against a random-init control, and validate probe protocols on a random encoder.